

Creating and Managing Users in Active Directory

An Easy-to-Follow Guide for Beginners

Step 1: Setting Up Your Virtual Environment

To create a **Domain Controller (DC)** and a **Client Machine**, you need virtualization software such as **VirtualBox**, **VMware Workstation**, or **Hyper-V**.

Download and Install VirtualBox (Recommended for Beginners)

1. Go to the [VirtualBox website](https://www.virtualbox.org/).
2. Download and install **VirtualBox** for your operating system.
3. Download the **Windows Server 2019 ISO** and **Windows 10 ISO** (available through Microsoft Evaluation Center).

Create a Virtual Machine for Your Domain Controller (DC)

1. Open VirtualBox and click **New** to create a virtual machine.
2. Name it "**Windows Server 2019 DC**" and set the following configurations:
 - **RAM:** At least **4GB** (recommended: 8GB).
 - **Hard Drive:** **50GB+ (Dynamically Allocated)**.
 - **Network Adapter:** Set to **Internal Network** (so it can communicate with the client machine).
3. Install **Windows Server 2019** using the downloaded ISO.
4. Once installed, configure the server as a **Domain Controller** using **Active Directory Domain Services (AD DS)**.

Create a Virtual Machine for Your Client Machine (Windows 10)

1. In VirtualBox, create a new VM and name it "**Windows 10 Client**".
 2. Set the following configurations:
 - **RAM:** At least **2GB** (recommended: 4GB).
 - **Hard Drive:** **30GB+ (Dynamically Allocated)**.
 - **Network Adapter:** **Internal Network** (same as the DC).
 3. Install **Windows 10** using the ISO.
 4. Once installed, **join it to the domain** created in Windows Server 2019.
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Step 2: Download and Prepare the User Creation Script

1. On your **Domain Controller (DC)**, open **Internet Explorer**.
 2. Download the **PowerShell user creation script** (provided by your instructor or IT department).
 3. Save the script to your **Desktop** and extract it.
 4. Inside the extracted folder, locate:
 - **PowerShell script** (for user creation).
 - **names.txt** (list of users to be created).
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Step 3: Modify the User List

1. Open "**names.txt**" in Notepad.
 2. Add your name at the top of the list (e.g., "Gustavo ColonRivera").
 3. Save and close the file.
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Step 4: Open PowerShell as Administrator

1. Click **Start**, search for **PowerShell ISE**, and **right-click** to select "**Run as Administrator**".

In PowerShell, **enable script execution**:

powershell

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Set-ExecutionPolicy Unrestricted

- 2.
3. Press **Enter** and confirm by typing **Y (Yes to All)**.

Step 5: Understanding the Script

The script will:

- ✓ Read names from "**names.txt**" and create user accounts.
- ✓ Set a **default password** for all accounts.
- ✓ Place the users inside an **Organizational Unit (OU)** called "Users".
- ✓ Ensure all accounts are enabled after creation.

Step 6: Run the Script to Create Users

In PowerShell, navigate to the script's directory:

powershell

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cd C:\Users\a-gcolonrivera\Desktop\UserCreationScript

- 1.

Run the script:

powershell

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```
.\CreateUsers.ps1
```

2.

3. If prompted, select **Run Once** to execute the script.

The script will display messages like:

sql

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```
Creating user: a-gcolonrivera
```

```
Creating user: jsmith
```

```
Creating user: mrodriguez
```

4.

Step 7: Verify Users in Active Directory

1. Open **Active Directory Users and Computers (ADUC)**:
 - Click **Start** → **Administrative Tools** → **Active Directory Users and Computers**.
 2. Navigate to the **"Users"** Organizational Unit (OU).
 3. Search for your account (**gcolonrivera**) using the **Find** option.
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Step 8: Test Login with a Domain Account

1. On your **Windows 10** client machine:
 - Click **Other User** at the login screen.
 - Sign in using your **Active Directory** username:
 - **Username:** a-gcolonrivera
 - **Password:** Password1 (default password).

2. If login is successful, your account is now part of the **Active Directory domain!**
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Conclusion

You have successfully:

- ✓ Set up a **Windows Server 2019 Domain Controller** in **VirtualBox**.
- ✓ Created a **Windows 10 client machine** and joined it to the domain.
- ✓ Used **PowerShell** to create and manage **Active Directory** users.
- ✓ Logged in as a domain user to verify the setup.